

Order Statistics for Discrete Random Variables

- Let $f(x) = 1/6$, $x = 1, 2, 3, 4, 5, 6$, zero elsewhere, be the pmf of a distribution of the discrete type. Show that the pmf of the smallest observation of a random sample of size 5 from this distribution is

$$g_1(y_1) = \left(\frac{7-y_1}{6}\right)^5 - \left(\frac{6-y_1}{6}\right)^5, y_1 = 1, 2, 3, 4, 5, 6.$$

References

- [1] *Introduction to Mathematical Statistics* by Hogg, McKean, Craig
- [2] *Probability and Statistics: The Science of Uncertainty* by Evans and Rosenthal
- [3] *A First Course in Probability* by Ross
- [4] *John E. Freund's Mathematical Statistics with Applications* by Irwin Miller Marylees Miller